Berka Database

Part 2: Loan Analysis

1. Project Summary

This project focuses on the loan problem: how to make more profit with less risk. Using known previous loan outcomes from the Berka database, we study what information, available *before* a loan is granted, tells us whether a client will repay it or default. This is treated as a binary classification problem between good and bad outcomes.

For each client, we looked at three types of information that may affect the loan outcome: Loan Characteristics, Static Account Details, and Historical Account Behavior. For each type, we extracted the required or useful variables that represent it. Then we analyze the extracted data by looking for patterns through visualizations and by applying the logistic regression method.

The results show that a logistic regression with four variables—loan duration, capacity, R(consistency), and North Bohemia region—reaches a cross-validated AUC of about 0.83. Longer loans and lower consistency (higher R) raise the risk of default, while more room above the payment lowers it.

279	**27%**	**6 → 4**	**0.83**
loans analyzed	of them bad	features tested → kept	cross-validated AUC

2. Data & Methodology

The data comes from the Berka dataset (LOANS table), linked to other tables to extract the required variables (see the Data folder from the previous study). All work was done in Python (Colab).

We have 682 loans:

Status	Meaning	Count	Used as
A	Finished, paid	203	Good (0)
B	Finished, not paid	31	Bad (1)
C	Running, in good standing	403	Excluded (outcome unknown)
D	Running, in debt	45	Bad (1)

So the final dataset has **279 loans, 27% of them bad**.

For a client asking for a loan. The bank's challenge is to judge whether the client can truly sustain the payments (paying a specific amount each month for a specific number of months), relying strictly on information available before the loan is granted.

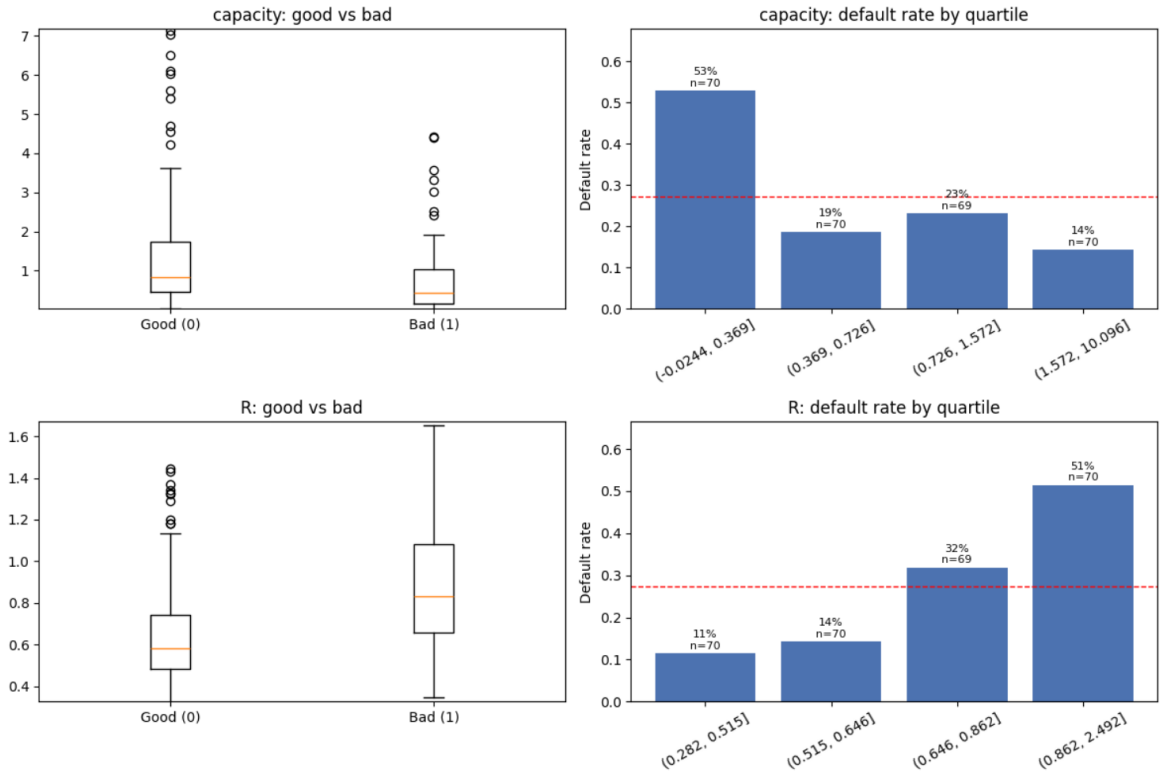
So we are looking for information that supports stability, consistency, and ability. Because unpredictable life events—such as job changes—are absent from the dataset, the focus was to identify stable, reliable patterns within the available account data to guide the lending decision.

Information type	Question	Feature	Definition
Static Account Details	What environment or circumstances is he in?	Region	Region of the account owner
Static Account Details	What type of income does he bring?	Salary_Flag	1 if the account receives regular salary-type transfers, otherwise 0
Static Account Details	How long have we known him?	Account_Age	Months from account opening to the loan date
Loan Characteristics	For how long must he keep it up?	Loan_Duration	Number of months of the loan
Historical Account Behavior	How much can he set aside each month?	Capacity	Average monthly net (money in minus money out) divided by the monthly loan payment
Historical Account Behavior	How steady is that net money over months?	R (consistency)	Standard deviation of the monthly net divided by the average monthly income (for standardization)

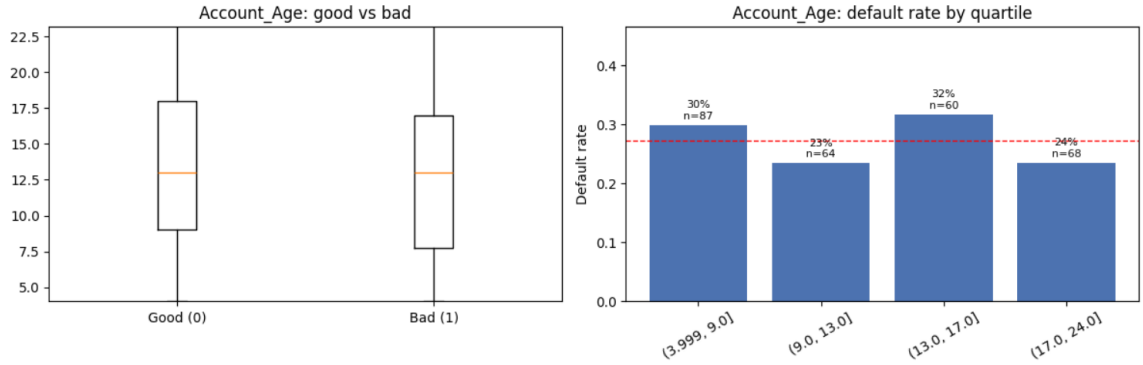
- Only information from before the loan date is used.
- **Capacity** is the level. A value of 1 means the client keeps, on average, exactly the amount of the payment. Above 1 is better, below 1 is worse.
- **R (Consistency)** is the stability. Some clients leave a lot of net money one month and little in the next. The higher R is, the less reliable the flow.

3. Exploration

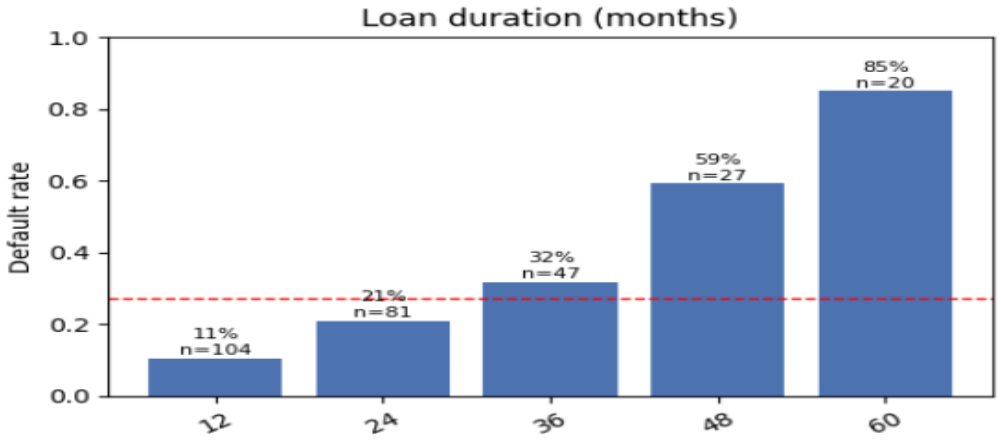
Before modeling, each feature was compared across good and bad loans, using the default rate within groups.



- **Capacity:** "Good" borrowers generally have higher capacity numbers than "Bad" borrowers. The lowest quartile is clearly riskier (53%) than the other three (14–23%).
- **R:** "Bad" borrowers tend to have higher R values overall. As R increases, default risk jumps from 11% to 51%.



- **Account age:** "Good" and "Bad" borrowers have very similar account-age distributions, and default rates fluctuate up and down across the quartiles without a clear trend.



- **Loan duration:** the default rate jumps dramatically from 11% for 12-month loans to 85% for 60-month loans.



- **Salary Flag**: People with a salary (1) have a lower default risk (17%) compared to those without a salary (0), who have a higher risk (31%).
- **Region:** Default risk varies by location, ranging from a low of 5% in North Bohemia up to a high of 35% in North Moravia.

4. Modeling

A logistic regression estimates the probability of default from a weighted combination of the features. We standardized the variables so that each effect is measured for a typical-size increase (one standard deviation) and effects can be compared. Region was split into 0/1 variables, with *East Bohemia* as the reference.

We measured performance with 5-fold cross-validation, so every loan is scored by a model that never saw it, and report three standard measures:

- **AUC (The Sorting Score):** how well the model ranks bad loans above good ones. 0.5 is a coin flip and 1.0 is a perfect separation.
- **Accuracy (The Overall Score):** the share of loans classified correctly. Since 73% of the loans are good, a model that always says "good" already scores 0.728 (our baseline).
- **F1 Score (The Bad Loan Balance):** how well the model finds the bad loans, balancing the bad loans it misses against its false alarms.

The model was built in two stages:

Model	Parameters	Variables	AUC	Accuracy	F1	AIC	BIC
Stage 1: all features	12	Loan_Duration, salary_Flag, capacity, R, Region, Account_Age	0.814	0.807	0.584	246.059	293.265
Stage 2: significant features	4	Loan_Duration, capacity, R, Region_north Bohemia	0.833	0.799	0.584	236.276	254.432
Baseline	–	–	0.500	0.728	0.000	–	–

Both models are clearly better than the baseline. Removing the extra features changed nothing important: accuracy and F1 stay essentially the same (differences below 0.01), AUC improves, and the simpler model has the lower AIC and BIC (lower is better). A likelihood-ratio test confirms that the eight removed parameters add nothing significant together ($\chi^2 = 6.22$, df = 8, $p = 0.623$). We kept the simpler model.

Final model

Variable	Coefficient	Odds ratio (95% interval)	p-value	Effect
Loan duration	+ 1.006	2.733 (1.968 – 3.797)	0.000	Raises risk
R	+ 0.871	2.389 (1.674 – 3.409)	0.000	Raises risk
Capacity	− 0.608	0.544 (0.320 – 0.925)	0.025	Lowers risk
Region_north Bohemia	− 2.900	0.055 (0.005 – 0.605)	0.018	Lowers risk

What the final model gets right and wrong (Confusion matrix):

	Predicted good	Predicted bad	Total
Actually good	184	19	203
Actually bad	37	39	76
Total	221	58	279

Out of 203 good loans, the model correctly accepts 184 (91%). Out of 76 bad loans, it catches 39 (51%). When it flags a loan as bad, it is right 39 times out of 58 (67%).

5. Key Observations

The final model is simple, but it gives a clear picture of which loans are risky and why.

- **Loan duration is the strongest risk factor:** each extra standard deviation of duration multiplies the odds of default by about 2.7. Longer loans increase default risk dramatically because a longer repayment period exposes the client to more life uncertainties, so a long term alone is a serious warning sign.
- **Both the level and the stability of a client's net money matter:** higher capacity lowers risk (odds ratio 0.54, about 46% lower odds), while a less consistent flow (higher R) raises it (odds ratio 2.39). The safest client keeps more than the payment each month, and does so steadily.
- **Salary flag and account age add little:** salary-type income looked protective on its own (17% vs 31% default), but the effect disappears once the other variables are included ($p = 0.597$). Account age shows no clear pattern ($p = 0.335$). Their information is most likely already captured by capacity and R.
- **Region matters only for North Bohemia, and it should be treated with caution:** its default rate is far lower (5%), but the wide interval (0.005–0.605) shows the estimate is uncertain, probably because of the small number of loans. It is a hint, not a lending rule.
- **The model works as a screening tool, not a final decision-maker:** a cross-validated AUC of 0.83 and an accuracy of 0.80 (baseline 0.728) show it ranks risky loans well. It accepts 91% of good loans and catches 51% of bad ones, and when it flags a loan it is right two times out of three (67%). Still, 37 of the 76 bad loans pass through.
- **Simpler models give better results**: Removing 8 weak variables left us with a 4-variable model that achieved a higher AUC (0.833 vs 0.814) and lower AIC/BIC, proving that dropping unnecessary noise improves predictions.

In practice

To get more profit with less risk, the bank should be careful with long terms, check that the client's monthly net money comfortably covers the payment, and look for a steady flow rather than a high average. A short loan to a client with high capacity and low R is the safest; a long loan to a client with an irregular flow is the riskiest.